

AGREEMENT COEFFICIENTS WITH MISSING RATERS

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ABSTRACT. Missing data is a pervasive challenge in studies of inter-rater agreement, yet common agreement coefficients, such as Conger’s kappa, Fleiss’ kappa, and the Brennan–Prediger coefficient, are often applied without clear justification when ratings are incomplete. We develop a general framework for handling missing data in agreement analysis that supports both categorical and continuous responses, allows arbitrary disagreement functions, and hold provided the data is pairwise missing completely at random. Then we characterize when pairwise-available estimators remain consistent, derive their correct standard errors, and identify settings where conventional formulas fail. We then propose reweighted kappa estimators that remain valid under substantially weaker assumptions. Asymptotic covariance matrices enable joint inference across coefficients, and simulations verify the theoretical claims. The methods are implemented in an R package `misskappa` designed for applied use.

1. INTRODUCTION

A chance-corrected agreement coefficient is any coefficient on the form $\kappa_x = 1 - \frac{d}{d_x}$ where d is a measure of disagreement and d_x a measure of chance disagreement. Let X^* be a random vector with R components. The vector contains all rating data, including potentially unobserved ratings. Define disagreement as

$$(1.1) \quad d = \binom{R}{2}^{-1} \sum_{j < k} E[l(X_j^*, X_k^*)],$$

where $l(x, y) \geq 0$ is a symmetric loss function satisfying $l(x, x) = 0$. The by far most common are the nominal ($l(x, y) = 1[x \neq y]$), absolute value ($l(x, y) = |x - y|$), and quadratic ($l(x, y) = (x - y)^2$) losses.

We cover three variants of chance disagreement, the Fleiss-type [Fleiss, 1971], Cohen-type [Cohen, 1968], and Brennan–Prediger-type [Brennan and Prediger, 1981]. These are defined as

$$(1.2) \quad d_F = R^{-2} \sum_{j, k} E[l(X_{ij}^*, X_{i'k}^*)],$$

$$(1.3) \quad d_C = \binom{R}{2}^{-1} \sum_{j < k} E[l(X_{ij}^*, X_{i'k}^*)],$$

$$(1.4) \quad d_{BP} = C^{-2} \sum_{a, b} l(a, b),$$

where $X_{i\cdot}^*$ and $X_{i'\cdot}^*$ are independent samples from the same distribution. In the case of Brennan–Prediger we must assume that X^* takes on $C < \infty$ categorical values. Interpretations of these chance disagreements are well-known and can be found elsewhere.

For complete data, the kappa coefficients are estimated using the natural sample counterparts. Their standard errors may be found using either multinomial theory or U -statistics [Moss, 2024].

Our task is to estimate and do inference on agreement coefficients in the presence of missing or incomplete data. When dealing with incomplete data we do not observe all ratings in X^* . We formalize this by defining

$$X_j = \begin{cases} X_j^* & M_j = 1, \\ \text{NA} & \text{otherwise.} \end{cases}$$

where M is the missingness indicator and NA denotes a missing value. We naturally assume that X_j^* cannot take on the value NA.

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2. HÁJEK-TYPE INVERSE PROBABILITY-WEIGHTED ESTIMATOR

We begin by describing an inverse probability-weighted estimator of the Hájek-type.

2.1. Definition. Define the observation probabilities $\pi_j = P(M_j = 1)$ and $\pi_{jk} = P(M_j M_k = 1)$. Then define the natural sample estimators $\hat{\pi}_j = \hat{P}(M_j = 1)$ and $\hat{\pi}_{jk} = \hat{P}(M_j M_k = 1)$, with $\hat{P}$ being the empirical probabilities.

The *Hájek-type inverse probability-weighted estimator* is $\hat{\kappa}_X^{IPW} = 1 - \hat{d}^{IPW} / \hat{d}_C^{IPW}$, where we define

$$\hat{d}^{IPW} = \frac{\sum_{i=1}^n \sum_{j < k} l(X_{ij}, X_{ik}) M_{ij} M_{ik} \hat{\pi}_{jk}^{-1}}{\sum_{i=1}^n \sum_{j < k} M_{ij} M_{ik} \hat{\pi}_{jk}^{-1}}, \quad \hat{d}_C^{IPW} = \frac{\sum_{i,i'} \sum_{j < k} l(X_{ij}, X_{i'k}) M_{ij} M_{i'k} \hat{\pi}_j^{-1} \hat{\pi}_k^{-1}}{\sum_{i,i'} \sum_{j < k} M_{ij} M_{i'k} \hat{\pi}_j^{-1} \hat{\pi}_k^{-1}}$$

with $\hat{d}_F^{IPW}$ being defined analogously to $\hat{d}_C^{IPW}$, with summation over all j, k . This is the natural extension of the Hájek estimator to the agreement setting.

It follows from the strong law of large numbers, Slutsky's theorem, and the continuous mapping theorem [see, e.g., [van der Vaart, 2000](#)] that

$$\hat{d}^{IPW} \xrightarrow{a.s.} 2.1 \binom{R}{2}^{-1} \sum_{j < k} \pi_{jk}^{-1} E[l(X_{1j}, X_{1k}) M_{1j} M_{1k}], \quad \hat{d}_C^{IPW} \xrightarrow{a.s.} \binom{R}{2}^{-1} \sum_{j < k} \pi_j^{-1} \pi_k^{-1} E[l(X_{1j}, X_{2k}) M_{1j} M_{2k}],$$

with a similar limit for $\hat{d}_F^{IPW}$. An obvious sufficient condition for $\hat{\kappa}_X^{IPW}$ to be consistent is that $\pi_{jk}^{-1} E[l(X_{ij}, X_{ik}) M_{ij} M_{ik}] = E[l(X_{ij}^*, X_{ik}^*)]$ and $\pi_j^{-1} \pi_k^{-1} E[l(X_{ij}, X_{i'k}) M_{ij} M_{i'k}] = E[l(X_{ij}^*, X_{i'k}^*)]$, and these conditions hold when the data is missing completely at random. We also require that $d_X > 0$ for κ_X to be defined, and that all $\pi_{jk} > 0$, so all rater combinations are counted.

Proposition 1. *The estimators $\hat{\kappa}_X^{IPW}$ are strongly consistent for all $X \in \{C, F, BP\}$ when the data is missing completely at random, exchangeable raters or not, provided that $d_X > 0$ and all $\pi_{jk} > 0$.*

Proof. That $\pi_{jk}^{-1} E[l(X_{ij}^*, X_{ik}^*) M_{ij} M_{ik}] = E[l(X_{ij}^*, X_{ik}^*)]$ is equivalent to $E[l(X_{ij}^*, X_{ik}^*)] = E[l(X_{ij}^*, X_{ik}^*) | M_{ij} M_{ik} = 1]$, which holds true when data is missing completely at random. The same reasoning holds for $\pi_j^{-1} \pi_k^{-1} E[l(X_{ij}^*, X_{i'k}^*) M_{ij} M_{i'k}] = E[l(X_{ij}^*, X_{i'k}^*)]$ as well. $\square$

Motivation sketch. Proposition 1 tells us that $\hat{\kappa}_X^{IPW}$ is strongly consistent. This is nice, but we usually desire our estimators to be efficient. It is well-known from semi-parametric efficiency theory that this estimator is not efficient [[Tsiatis, 2006](#)]. Constructing efficient estimators either require complex semi-parametric machinery for general data or iterative methods such as the EM-algorithm for categorical data. We intend to explore the EM-algorithm in a future paper, but will remark that semiparametric methods for efficiency gains is likely to backfire, and the complexity is not worth the investment in our situation.

Our motivation for the Hájek-type estimator cannot be based on efficiency directly, but we can make a weaker case for it. If we construct the population values κ_X^{IPW} using the limits in equation 2.1, the plugin-estimator $\hat{\kappa}_X^{IPW}$ is trivially efficient from standard theory. Moreover, it is consistent under reasonable assumptions that are weaker than missing completely at random. This is more than what we can say of its competitors

2.2. On inference. The estimators $\hat{\kappa}_X^{IPW}$ are asymptotically normal under the mild condition that the random values $l(X_{ij}^*, X_{ik}^*) M_{ij} M_{ik} \hat{\pi}_{jk}^{-1}$ have finite second moments. For each $X \in \{C, F, BP\}$, we have that

$$\sqrt{n}(\hat{\kappa}_X^{IPW} - \kappa_X^{IPW}) \xrightarrow{d} N(0, \sigma_X^2),$$

where κ_X^{IPW} is the almost sure limit of $\hat{\kappa}_X^{IPW}$, which equals κ_X under the conditions of Proposition 1. The asymptotic variance σ_X^2 , can be expressed using an (efficient) influence function in the observed-data model, and we estimate it by plugging in estimated influence functions.

$$(2.2) \quad \phi_{\kappa_X} = -d_X^{-1} \phi_d + d d_X^{-2} \phi_{d_X}.$$

To make this operational, write the IPW estimators as ratios. For the observed disagreement, $\hat{d}^{IPW} = \hat{\psi}_{dN} / \hat{\psi}_{dD}$ where

$$\hat{\psi}_{dN} = \frac{1}{n} \sum_{i=1}^n h_{dN,i}, \quad h_{dN,i} = \sum_{j < k} l(X_{ij}, X_{ik}) M_{ij} M_{ik} \hat{\pi}_{jk}^{-1}, \quad \hat{\psi}_{dD} = \frac{1}{n} \sum_{i=1}^n h_{dD,i}, \quad h_{dD,i} = \sum_{j < k} M_{ij} M_{ik} \hat{\pi}_{jk}^{-1}.$$

Then

$$\hat{\phi}_{d,i} = \hat{\psi}_{dD}^{-1} (\hat{\phi}_{dN,i} - \hat{d}^{IPW} \hat{\phi}_{dD,i}), \quad \hat{\phi}_{dN,i} = h_{dN,i} - \hat{\psi}_{dN}, \quad \hat{\phi}_{dD,i} = h_{dD,i} - \hat{\psi}_{dD}.$$

Similarly, for $X \in \{C, F\}$ write $\hat{d}_X^{IPW} = \hat{\psi}_{XN} / \hat{\psi}_{XD}$ with $\hat{\psi}_{XN}, \hat{\psi}_{XD}$ defined by the corresponding double sums in the definition of $\hat{d}_X^{IPW}$. Let K_{XN} and K_{XD} denote the associated kernels, and define $Z_i = (X_i, M_i)$ so that

$$\hat{\psi}_{X\bullet} = \frac{1}{n^2} \sum_{i,i'} K_{X\bullet}((X_i, M_i), (X_{i'}, M_{i'})), \quad X\bullet \in \{XN, XD\}.$$

Estimate their influence functions by the usual V-statistic projections:

$$\hat{\phi}_{X\bullet,i} = (\hat{g}_{X\bullet,1,i} - \hat{\psi}_{X\bullet}) + (\hat{g}_{X\bullet,2,i} - \hat{\psi}_{X\bullet}), \quad \hat{g}_{X\bullet,1,i} = \frac{1}{n} \sum_{i'} K_{X\bullet}(Z_i, Z_{i'}), \quad \hat{g}_{X\bullet,2,i} = \frac{1}{n} \sum_{i'} K_{X\bullet}(Z_{i'}, Z_i),$$

and then set

$$\hat{\phi}_{dX,i} = \hat{\psi}_{XD}^{-1} (\hat{\phi}_{XN,i} - \hat{d}_X^{IPW} \hat{\phi}_{XD,i}).$$

For $X = BP, d_X = d_{BP}$ is constant, so $\phi_{dX} = 0$. A consistent estimator of σ_X^2 is

$$\hat{\sigma}_X^2 = \frac{1}{n} \sum_{i=1}^n \hat{\phi}_{\kappa_X,i}^2,$$

and the same construction yields a joint asymptotic covariance matrix by replacing the square with an outer product. The same influence-function formalism applies to the available-case and Gwet-estimators, introduced below, by substituting their corresponding weights in the definitions of $\hat{d}$ and $\hat{d}_X$.

2.3. Computation.

2.4. Illustrations.

3. OTHER ESTIMATORS

3.1. Available-case estimator. The available-case (pairwise complete) estimator is a natural baseline when ratings are incomplete. It uses only the observed rater pairs and treats all observed pairs equally. Contrary to some common claims, the default missing-data behavior in the popular R package `irrCAC` is based on a Gwet-style reweighting; see the next subsection. We may define the available case Conger's kappa estimator as $\hat{\kappa}_C^{AC} = 1 - \hat{d}^{AC} / \hat{d}_C^{AC}$, where

$$\hat{d}^{AC} = \frac{\sum_{i=1}^n \sum_{j < k} l(X_{ij}^*, X_{ik}^*) M_{ij} M_{ik}}{\sum_{i=1}^n \sum_{j < k} M_{ij} M_{ik}}, \quad \hat{d}_C^{AC} = \frac{\sum_{i,i'} \sum_{j < k} l(X_{ij}^*, X_{i'k}^*) M_{ij} M_{i'k}}{\sum_{i,i'} \sum_{j < k} M_{ij} M_{i'k}}.$$

Observe that $l(X_{ij}^*, X_{ik}^*) M_{ij} M_{ik} = l(X_{ij}, X_{ik})$ if we extend the loss l to satisfy $l(x, y) = 0$ when x or y is NA.

Definition 1. Let X^* be a random vector of size R and X be the observed incomplete data. The components of X are missing *uniformly at random* if the component-wise missingness probabilities are constant (i.e., $P(M_i) = \pi$ for all i) and M_i and M_j are independent.

Proposition 2. *The available case estimator is strongly consistent if either*

- (1) *The raters are exchangeable and data is missing completely at random.*
- (2) *The raters are not exchangeable and data is missing uniformly at random.*

The estimator is in general not consistent when data is missing at random, even when raters are exchangeable.

3.2. Gwet's estimator. Gwet's estimator is a reweighted variant that keeps the observed disagreement component but changes the chance disagreement component to compensate for unequal rater observation probabilities. In our notation we define, for each $X \in \{C, F, BP\}$, $\hat{\kappa}_X^G = 1 - \hat{d}^G / \hat{d}_X^G$, where the observed disagreement is the same as the available-case estimator,

$$\hat{d}^G = \hat{d}^{AC} = \frac{\sum_{i=1}^n \sum_{j < k} l(X_{ij}^*, X_{ik}^*) M_{ij} M_{ik}}{\sum_{i=1}^n \sum_{j < k} M_{ij} M_{ik}}.$$

For the chance disagreement we use empirical rater observation probabilities $\hat{\pi}_j = \hat{P}(M_j = 1)$ and define

$$\hat{d}_C^G = \frac{\sum_{i,i'} \sum_{j < k} l(X_{ij}^*, X_{i'k}^*) M_{ij} M_{i'k} \hat{\pi}_j^{-1} \hat{\pi}_k^{-1}}{\sum_{i,i'} \sum_{j < k} M_{ij} M_{i'k} \hat{\pi}_j^{-1} \hat{\pi}_k^{-1}}, \quad \hat{d}_F^G = \frac{\sum_{i,i'} \sum_{j,k} l(X_{ij}^*, X_{i'k}^*) M_{ij} M_{i'k} \hat{\pi}_j^{-1} \hat{\pi}_k^{-1}}{\sum_{i,i'} \sum_{j,k} M_{ij} M_{i'k} \hat{\pi}_j^{-1} \hat{\pi}_k^{-1}}.$$

For $X = BP$ we keep d_{BP} as in equation 1.4.

4. SIMULATIONS

5. DATA FOR FLEISS' KAPPA ON COUNT FORM

Until now we have discussed estimators that require data on the form $X^* \in \mathbb{R}^J$

6. THE R PACKAGE `misskappa`

7. CONCLUDING REMARKS

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APPENDIX A. APPENDIX: PROOFS

A.1. Proof of Proposition 2.

of Proposition 2. We sketch the argument for $\hat{\kappa}_C^{AC} = 1 - \hat{d}^{AC} / \hat{d}_C^{AC}$; the same reasoning applies to the corresponding Fleiss and Brennan–Prediger versions. Define

$$C_{jk}(x^*) = P(M_j M_k = 1 \mid X^* = x^*), \quad C_j(x^*) = P(M_j = 1 \mid X^* = x^*).$$

Then

$$E[l(X_j^*, X_k^*) M_j M_k] = E[l(X_j^*, X_k^*) C_{jk}(X^*)], \quad E[M_j M_k] = E[C_{jk}(X^*)].$$

By the law of large numbers and the continuous mapping theorem,

$$\hat{d}^{AC} \xrightarrow{a.s.} \frac{\sum_{j < k} E[l(X_j^*, X_k^*) C_{jk}(X^*)]}{\sum_{j < k} E[C_{jk}(X^*)]}.$$

Next, let Y^* be an independent copy of X^* . A similar calculation yields

$$\hat{d}_C^{AC} \xrightarrow{a.s.} \frac{\sum_{j < k} E[l(X_j^*, Y_k^*) C_j(X^*) C_k(Y^*)]}{\sum_{j < k} E[C_j(X^*)] E[C_k(Y^*)]},$$

where the $i = i'$ terms are negligible relative to the $i \neq i'$ terms. Therefore $\hat{\kappa}_C^{AC}$ converges to the corresponding ratio-of-limits. If the raters are exchangeable and the data are missing completely

at random, then $C_{jk}(X^*) = \pi_{jk}$ and $C_j(X^*) = \pi_j$ are constants, while exchangeability implies $E[l(X_j^*, X_k^*)]$ and $E[l(X_j^*, Y_k^*)]$ do not depend on the indices. Hence the constants π_{jk} and $\pi_j\pi_k$ cancel in the limits and we obtain $\hat{d}^{AC} \rightarrow d$ and $\hat{d}_C^{AC} \rightarrow d_C$, so $\hat{\kappa}_C^{AC} \rightarrow \kappa_C$. If raters are not exchangeable but the missingness is uniformly at random (Definition 1), then the observation probabilities are constant across indices, so the limits again reduce to the unweighted averages defining d and d_C . Finally, the displayed limits show why inconsistency is typical under more general mechanisms: if $C_{jk}(X^*)$ or $C_j(X^*)$ depends on X^* in a way that is correlated with $l(X_j^*, X_k^*)$ or $l(X_j^*, Y_k^*)$, then the probability limits target reweighted versions of d and d_C rather than the desired parameters. $\square$

Remark 1. The same limit calculation can be carried out for Gwet's estimator as defined in the main text by replacing the chance-disagreement sums with their $\hat{\pi}_j^{-1}\hat{\pi}_k^{-1}$ reweighted versions. Under MCAR the reweighting cancels (since $C_j(X^*) = \pi_j$), but under more general mechanisms where $C_j(X^*)$ depends on X^* , the probability limit of the chance term is generally a reweighted target and need not equal d_C or d_F .

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